EECE5576: Wireless Communication Systems: Homework Assignment 1

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Submission Instructions:
Submit on Canvas:

- A report in electronic format (.pdf) named 'LastName_HW1.pdf'
- The MATLAB files (.m) you used, properly named (e.g. LastName_HW1a.m, LastName_HW1b.m, LastName_HW1c.m, LastName_HW1d.m).

Files provided with this assignment: `HW1_dataset_A.csv`, `HW1_dataset_B.csv` (equivalently `HW1_datasets.mat`), and `HW1_partD_AI_draft.m`.

Use of AI tools in this assignment. This assignment has two regimes, and they are graded differently.

- **Parts (a), (b), (c) and (d.1): AI tools are permitted** as a resource, on the same terms as any other resource. What is assessed here is not the production of the code or the figure; it is whether you can *verify* what you submit. Every submission must therefore include the **Verification Appendix** described at the end of this assignment, whether or not you used an AI tool. All code comments must be written in your own words.
- **Part (d.2): AI tools are not permitted.** The skill being assessed in (d.2) *is* the audit. Using an AI tool to perform the audit defeats the purpose of the assessment.

Submitting output you cannot explain, including fabricated references or results you cannot justify, is an academic integrity violation and will be handled under the Academic Integrity policy in the syllabus. The instructor may ask any student for a short (3–5 minute) oral walkthrough of any part of this submission.

Model used throughout this assignment. We use the simple (log-distance) path loss model with a close-in reference distance d_0,

$$\overline{PL}_{dB}(d) = \overline{PL}_{0|dB}(d_0) + 10\,n\log_{10}\left(\frac{d}{d_0}\right), \qquad d \geq d_0 \geq d_R, \tag{1}$$

where n is the path loss exponent and $\overline{PL}_{0|dB}(d_0)$ is the path loss at the reference distance. In all parts, obtain $\overline{PL}_{0|dB}(d_0)$ from the **free-space path loss** model with isotropic antennas,

$$PL_{fs} = \left(\frac{4\pi d f_c}{c}\right)^2. \tag{2}$$

Unless stated otherwise, assume unity antenna gains ($G_t = G_r = 1$).

a) **(15 points) Distance dependence and the path loss exponent.**

Consider $f_c = 900$ MHz and reference distance $d_0 = 100$ m. Plot the path loss $\overline{PL}_{dB}(d)$ of Eq. (1) as a function of the distance d, for d from 100 m to 10 km. Use a logarithmic scale for the distance axis. Plot 3 lines, for $n = 2$, $n = 2.7$ and $n = 3.5$.

In your report:

i. Include the figure, properly labeled (axis labels with units, legend, grid).

ii. Report the numerical value of $\overline{PL}_{0|dB}(100 \text{ m})$ that you used, and show the hand calculation that produced it.

iii. Read the slope of each line off your own figure, in dB per decade of distance, and compare it with the values predicted by Eq. (1). Do the three measured slopes agree with the three predicted values?

iv. Using the table of typical path loss exponents from the lecture notes, state which propagation environment each of the three curves would represent, and explain what a change from $n = 2$ to $n = 3.5$ means physically.

b) **(15 points) Frequency dependence and the validity region.**

Now fix $n = 3.5$ and $d_0 = 100$ m, and plot $\overline{PL}_{dB}(d)$ as a function of d over the same distance range. Plot 3 lines, for $f_c = 900$ MHz, $f_c = 2.6$ GHz and $f_c = 28$ GHz.

In your report:

i. Include the figure, properly labeled.

ii. Describe the shape of the three curves relative to each other. Explain, from Eqs. (1) and (2), *why* they have this relationship, and compute analytically the offset in dB between the 900 MHz curve and each of the other two. Verify that your computed offsets match the figure.

iii. The model of Eq. (1) is only valid for $d \geq d_0 \geq d_R$, where d_R is the far-field (Fraunhofer) distance. Assuming a largest antenna dimension of $D = 0.5$ m, compute d_R at each of the three carrier frequencies. Is the choice $d_0 = 100$ m acceptable at all three frequencies? Justify the lower limit of the distance axis in your figure.

iv. In your figure, the only effect of frequency is a constant offset. Explain why that is a *modeling artifact* rather than a physical result, and name and briefly describe two physical propagation effects that become significant at 28 GHz and that this model cannot represent.

c) **(20 points) Nominal link budget and maximum range.**

A base station transmits with $P_{t|dB} = 43$ dBm at $f_c = 900$ MHz. The receiver requires a minimum received power level $P^*_{dB} = -100$ dBm to achieve the target SNR. Assume $G_t = G_r = 1$, take $d_0 = 100$ m, and consider only the path loss term (*nominal* link budget: no fading, no shadowing margin).

In your report:

i. Write the nominal link budget for this system and derive an analytical expression for the maximum range R, i.e. the largest distance at which $\overline{P}_{r|dB}(R) \geq P^*_{dB}$, as a function of n, $P_{t|dB}$, P^*_{dB}, $\overline{PL}_{0|dB}(d_0)$ and d_0.

ii. Plot R as a function of n for n from 2 to 5. Choose the axis scales so that the behavior is readable, and justify your choice.

iii. Report R for $n = 2$, $n = 2.7$ and $n = 3.5$ in a small table.

iv. Comment on the plausibility of the value you obtain for $n = 2$. Is that range physically meaningful for a terrestrial cellular base station? Which assumptions of this calculation break down first as R grows?

v. This is a *nominal* link budget: it uses the average path loss only. In a real deployment the received power at a given location fluctuates around this average. Without doing the calculation, state in which direction the required $P_{t|dB}$ would move once a shadowing (fading) margin is included, and explain why.

d) **(40 points) Fitting a path loss model to measurements, and auditing a proposed solution.**

Two sets of measurements are provided, each containing a transmitter-receiver distance d_i (in m) and the corresponding measured power loss $L_{dB}(d_i)$ (in dB):

	Dataset A (`HW1_dataset_A.csv`)	**Dataset B** (`HW1_dataset_B.csv`)
Environment	Outdoor, urban area	Indoor, office building, same floor
Carrier frequency	$f_c = 900$ MHz	$f_c = 2.4$ GHz
Reference distance	$d_0 = 100$ m	$d_0 = 1$ m
Measurements	$M = 60$ locations	$M = 40$ locations
Largest antenna dimension	$D = 0.5$ m (base station)	$D = 5$ cm (indoor access point)

Note that the two campaigns use very different antennas. Before using either dataset, check that the reference distance d_0 satisfies $d_0 \geq d_R$ for the corresponding D and f_c; recall from part (b) that $d_R = 2D^2/\lambda$ depends on the aperture as well as on the frequency.

(d.1) (20 points) Estimate the model parameters.

For *each* dataset, using $\overline{PL}_{0|dB}(d_0)$ obtained from free space at the corresponding d_0 and f_c:

i. Estimate the path loss exponent n that minimizes the mean square error between the measured loss and the model of Eq. (1), in dB units. Report the estimated $\hat{n}$ for each dataset and show the expression you minimized.

ii. Estimate the standard deviation σ_N of the residuals, using the sample estimator introduced in the lecture notes,

$$\sigma_N^2 = \frac{1}{M}\sum_{i=1}^{M}\left[L_{dB}(d_i) - \overline{PL}_{dB}(d_i)\right]^2 .$$

iii. For each dataset, plot the measurements as a scatter plot together with your fitted model, on a logarithmic distance axis. Two figures, or one figure with two subplots.

iv. Compare the two datasets. Which fit *looks* worse in the figures, and which one is the *less reliable estimate*? justify your answer using $\hat{n}$, σ_N and M. What does a larger σ_N tell you about the environment, as opposed to about the correctness of the model?

v. Compare your estimated exponents with the typical values from the lecture notes for these two environments. Are they consistent?

(d.2) (20 points) Audit of a proposed solution. *No AI tools may be used in this part.*

The file `HW1_partD_AI_draft.m` contains a proposed solution to part (d.1), produced with a generative AI tool, together with a short summary of its results written in the comments at the end of the file. The script runs without errors and produces figures that look plausible. **It is not correct.**

Your task is to audit it. In your report:

i. Identify the errors. For *each* error, state: (1) the line or block where it occurs; (2) what is wrong, in one or two sentences; (3) what the correct version is; (4) **the evidence that reveals it** (a physical argument, a unit check, a hand calculation, an inconsistency with the lecture notes, or a feature of the resulting figure or number). Item (4) carries the most weight.

ii. The number of errors is not given to you. Note also that not every unusual choice in the script is an error: if you believe a suspicious-looking line is in fact correct, say so and justify it. Flagging correct code as incorrect will lose points.

iii. State which of the errors you found would have been caught by simply looking at the output figure, and which ones would not. What does that tell you about how much you can trust plausible-looking output?

Verification Appendix (required, part of the 10 points below). At the end of your report, include a short appendix containing:

1. A statement of whether you used an AI tool for parts (a)–(d.1), which tool, and for what. If you did not use one, say so.
2. The prompts you used, verbatim, if any.
3. At least **three independent hand checks** of your own results, one per part (a), (b), (c). For example: the free-space path loss at d_0 computed in closed form; the dB-per-decade slope; one point of the link budget.
4. One thing that was wrong, or that you had to change, in the first version of your code or text, and how you detected it.

NOTE: (10 points) Write a clear and organized report, include the Verification Appendix, and be organized in your submission. Figures must be legible, with labeled axes, units and legends.